

Combat Robot Build Guide

Pure-RC Drive & Weapon System — ESC / Receiver / Transmitter Only (No Microcontroller)

This guide covers the fully RC-based control architecture for the drive and weapon systems: an FS-i6 transmitter and FS2A receiver controlling two bidirectional brushed ESCs (N20 drive motors) and one brushless ESC (B1806 weapon motor) directly, with no ESP32 or other microcontroller in the control loop.

It combines the wiring diagram, a verbose wiring walkthrough, transmitter programming steps, a pre-flight checklist, and a troubleshooting reference into a single document.

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1. System Overview

This build uses no microcontroller anywhere in the control path. The transmitter does all the work of combining stick movements into drive commands, and each ESC decodes its own receiver channel directly into motor speed and direction. This trades away software-defined behavior (custom failsafe logic, Wi-Fi/BLE control, weapon throttle shaping) for a simpler, fewer-parts system with one less thing that can crash or need reflashing mid-competition.

Three independent electronic speed controllers are used:

- **Two bidirectional brushed ESCs** — one per N20 drive motor, each capable of full forward/reverse.
- **One brushless ESC** (15A slim, 2S) — drives the B1806 2400KV weapon motor. This ESC is **throttle-only**: it has no reverse and no true center-neutral the way the drive ESCs do.

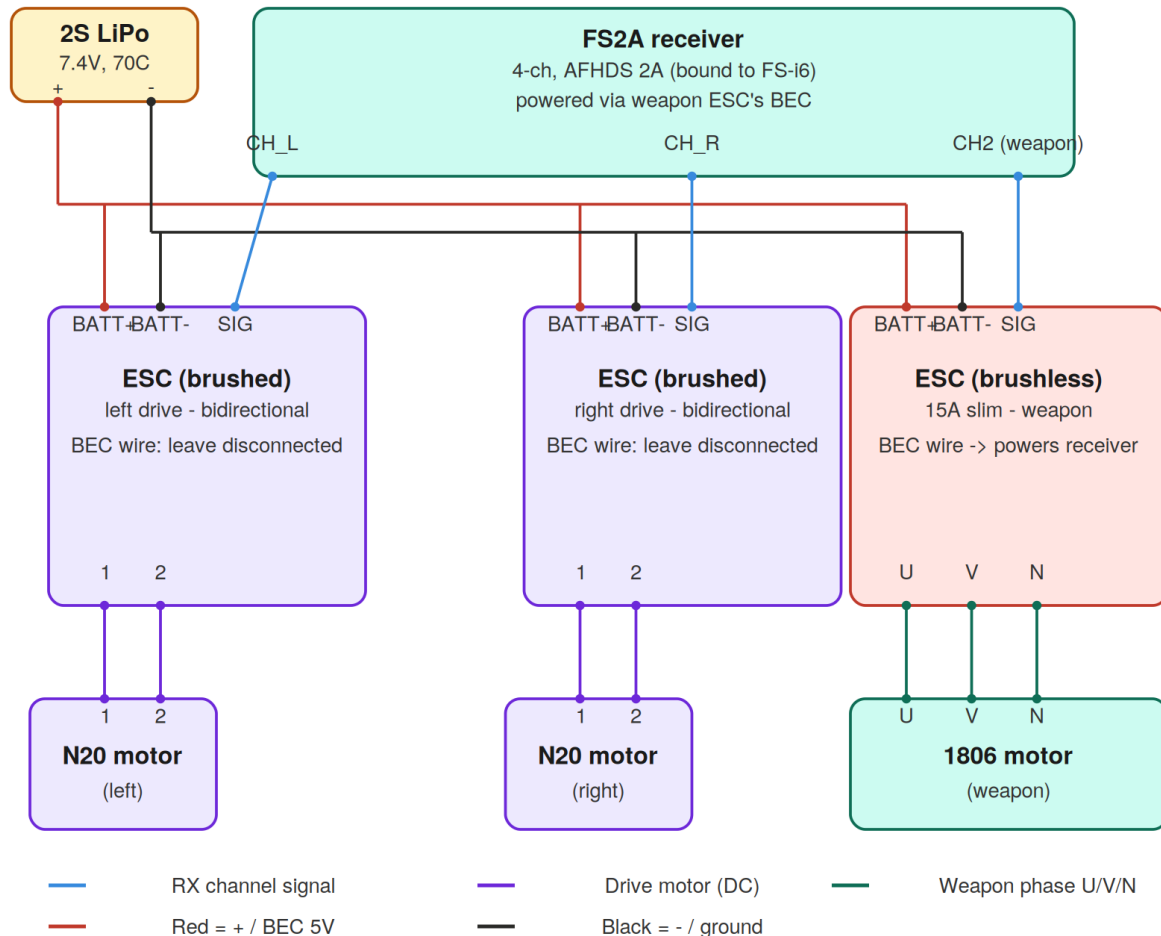
Note: The weapon ESC being throttle-only (not bidirectional) is the single most important distinction in this build. Treating it like a centered joystick channel is a common source of "weapon won't stop" problems — see the Troubleshooting section.

2. Parts List

Part	Spec	Qty
2S LiPo battery	7.4V nominal, 70C discharge rating	1
FlySky FS-i6 transmitter	6-channel, Mode 2 (US standard)	1
FS2A receiver	4-channel, AFHDS 2A, bound to FS-i6	1
Brushed bidirectional ESC	~3-5A continuous, reversible (car/tank style)	2
Brushless ESC	15-20A, 2S, slim profile, throttle-only	1
N20 gear motor	~500 RPM class @ 6V nominal	2
B1806 2400KV motor	Brushless, weapon/spinner use	1

Two brushed ESCs are listed as separate units; a single dual-channel bidirectional board (one board driving both N20 motors) works equally well and reduces part count — the wiring in this guide applies either way, just duplicate or split the "drive ESC" section as needed for your specific board.

3. Wiring Diagram



Full wiring diagram: battery, receiver, three ESCs, and their motors.

4. Detailed Wiring Instructions

4.1 Battery to ESCs

The battery's two leads (+/-) connect in parallel to all three ESCs' battery inputs. This is typically done with a small power-distribution board, a Y-harness, or by soldering the three sets of battery leads together at a single joint (heat-shrink each joint individually afterward). Keep polarity consistent across all three — double-check with a multimeter before the first power-up, since a reversed connection on even one ESC can destroy it instantly.

■ **Caution:** This battery is rated for 70C discharge, meaning it can supply an enormous amount of current very quickly. A short circuit at this stage is a real fire risk. Verify every connection before plugging in the battery, and never leave a connected battery unattended.

4.2 Receiver Connectors: the BEC Conflict Rule

Each ESC's 3-wire receiver harness carries three different things on three different wires: **white = signal (PWM)**, **red = BEC output (+5V)**, **black = ground**. All three ESCs plug into the receiver this way — but only **one** of those three red BEC wires should actually be connected.

Every ESC's BEC is a small voltage regulator generating 5V from the battery. If more than one ESC's red wire is connected to the receiver at the same time, two (or three) separate regulators end up fighting to hold the same power rail at slightly different voltages. This can cause erratic receiver behavior at best, and damage a regulator at worst.

- 1 Choose the weapon ESC as the one that powers the receiver — its BEC is well characterized and (per the manufacturer spec sheet) rated for a real 2A output, giving the most headroom.
- 2 Leave that ESC's red wire connected to its receiver channel plug as normal.
- 3 On **both** drive ESCs, clip back or insulate (heat-shrink or tape) the red BEC wire so it cannot make contact with anything. Signal (white) and ground (black) on those two connectors stay connected as normal.

4.3 Receiver Channel Assignments

With the transmitter mix described in Section 5, the receiver ends up with:

- **CH_L** (left drive, mixed output) → left drive ESC's signal wire
- **CH_R** (right drive, mixed output) → right drive ESC's signal wire
- **CH2** (unmixed, right stick vertical) → weapon ESC's signal wire

Note: The exact port numbers your mixed outputs land on depend on how the mix is configured in the transmitter menu — write down which physical receiver port ends up carrying which signal once it's set up, and label it on the receiver itself with tape.

4.4 Motor Connections

N20 drive motors: each N20 has two solder tabs with no polarity marking that matters at the ESC level — the ESC's electronics handle direction, not the motor. Solder each tab to one of the drive ESC's two motor-output wires. If a wheel spins backward once everything is powered up, swap that motor's two wires and re-solder.

B1806 weapon motor: this is a brushless motor with three phase wires, connecting to the weapon ESC's **U / V / N** solder pads (these are bare pads on a slim ESC, not a connector — see below). Order does not matter for function; if the weapon spins the wrong direction, swap any two of the three wires.

Note: U/V/N on a slim weapon ESC are typically bare solder pads, not a plug. Tin the wire ends and the pads before joining them, keep joints short so adjacent pads aren't bridged, and cover the finished joints with hot glue or heat-shrink — this connection sees real vibration and impact loads in combat.

5. Transmitter Programming (FlySky FS-i6)

The FS-i6's default channel-to-stick mapping (Mode 2, the US standard the radio ships in) is:

Channel	Stick / axis	Self-centering?
CH1 (Aileron)	Right stick, left/right	Yes
CH2 (Elevator)	Right stick, up/down	Yes
CH3 (Throttle)	Left stick, up/down	No — ratchet-style by default
CH4 (Rudder)	Left stick, left/right	Yes

5.1 Why a custom mix, not a preset

The FS-i6 ships with two named mixing presets, "Elevon" and "V-Tail", but neither produces "both drive axes on the left stick, weapon fully on the right stick": Elevon combines CH1+CH2 (both right-stick axes), and V-Tail combines CH2+CH4 (a cross-stick blend). What this build needs instead is a **custom Aux Channel Mix**:

- 1 Set **CH3 (throttle)** as the master input.
- 2 Set **CH4 (rudder)** as the slave input.
- 3 Create two mix instances — one at a positive rate, one inverted — outputting to two separate channels. These two outputs are your CH_L and CH_R drive signals.
- 4 Leave **CH2** completely unmixed and assign it to the weapon ESC.

5.2 The CH3 self-centering issue

■ **Caution:** Channel 3 (throttle) does not spring back to center by default on the FS-i6 — it's a ratchet-style stick, since a real airplane throttle doesn't need to auto-return to idle. Since this build's forward/reverse drive input rides on CH3, releasing the left stick would leave the bot still driving at whatever speed it was left at, rather than stopping. This should be fixed before the bot runs anywhere near people.

Two documented fixes:

- A physical hardware mod converting the CH3 stick mechanism to spring-centering.
- A firmware "secret menu" setting that allows the transmitter to arm with CH3 centered, instead of requiring it pulled to zero at power-on.

5.3 Receiver failsafe

Configure the receiver's failsafe behavior (via the transmitter's RX Setup submenu) to drive all channels to a safe/neutral value if the transmitter signal is lost. The default "hold last value" behavior on many FlySky receivers is not acceptable for a combat robot — a signal dropout should stop the bot, not freeze it at whatever speed it was moving.

5.4 Reference guide

Full button-by-button menu navigation for all of the above is covered in a dedicated community guide written specifically for combat robot use of the FS-i6:

- [FlySky FS-i6 Transmitter Programming for Combat Robots \(PDF\)](#)
- [Same guide, web version](#)

Note: This is a third-party guide written by a combat robot builder, not an official FlySky document — which is exactly why it's useful, since it translates the stock aircraft-oriented manual into combat-robot terms.

6. Pre-Flight Checklist

Run through this every time before connecting the main battery, and especially after any wiring change:

- 1 Transmitter powered on and confirmed bound to the receiver **before** connecting the battery.
- 2 Battery voltage checked — do not run a 2S pack below its safe discharge cutoff.
- 3 Visual check: no exposed/shorted wiring, no loose solder joints at the U/V/N pads or motor terminals.
- 4 Only one ESC's red BEC wire connected to the receiver — confirm the other two are insulated.
- 5 Weapon area clear of fingers, tools, and loose objects.
- 6 Connect the battery, then immediately test drive at low throttle before doing anything else.
- 7 With the bot lifted off the ground (wheels free-spinning), briefly test the weapon at low throttle and confirm it stops fully when the stick is released.
- 8 Turn the transmitter off while the bot is powered and confirm the failsafe behavior stops the bot, rather than freezing it at speed.

7. Troubleshooting Reference

Symptom	Likely Cause	Fix
Weapon spins but won't stop	Weapon ESC is throttle-only, not centered like a drive channel — treating it as self-centering is the most common cause	Confirm CH2 stick truly returns to its lowest position; check for a BEC conflict (see next row) which can cause erratic signal reception
Erratic control, receiver seems confused	More than one ESC's BEC (red wire) connected to the receiver at once	Disconnect/insulate all but one ESC's red wire
Weapon spins the wrong direction	Phase order	Swap any two of the three U/V/N wires at the ESC
A drive wheel spins the wrong direction	Motor wire polarity	Swap that motor's two wires at the ESC output
Robot turns opposite of stick input	Left/right channel assignment reversed	Swap the two drive ESCs' receiver channel plugs (simpler than re-doing the transmitter mix)
One side drives, other doesn't respond	Mix output channel mis-assigned, or wrong receiver port	Re-check which physical port the transmitter's custom mix is actually outputting to
Bot creeps forward/back at rest	Stick center/subtrim not adjusted, or missing deadzone	Adjust subtrim on CH3 so both drive outputs read neutral at rest

Symptom	Likely Cause	Fix
Nothing powers on	Battery wiring fault, or a BEC damaged from a prior conflict	Check battery connections first; if a BEC was previously run in conflict, test it in isolation before trusting it again
Receiver won't bind to transmitter	Wrong protocol/bind procedure, or receiver damaged	Confirm AFHDS 2A bind procedure for the FS2A specifically; re-check receiver power

8. Safety Notes

- The weapon ESC is throttle-only with no reverse — "off" only happens at the very bottom of its signal range. Verify this behavior on every power-up before trusting it.
- Always disconnect the main battery when the bot isn't actively being driven or tested.
- Keep hands, tools, and loose clothing clear of the weapon any time the battery is connected, even if the transmitter is off — a receiver failsafe fault or wiring issue could still energize it.
- Use eye and body protection during weapon testing, and test with the robot restrained or lifted clear of the ground for the first run after any wiring change.
- A 70C-rated battery can deliver very high current almost instantly — treat every wiring joint as a potential short-circuit and fire risk until verified.
- Re-run the full Pre-Flight Checklist after any rewiring, not just before a match.

Build guide compiled for the ESC/Receiver/Transmitter-only combat robot configuration. Cross-check part-specific details (exact BEC ratings, ESC calibration behavior) against your actual components, since manufacturers vary.